

- Null-space projector

$$N_\tau = I_7 - J^\top J^{+\top}, \quad \tau_{\text{null}} = \tau_d + \tau_\sigma$$

Projects secondary joint torques into the kinematic null space.

J : joint $\rightarrow$ EE velocities

J^+ : pseudoinverse

I_7 : 7×7 identity matrix

- Damping torque

$$\tau_d = -d_{\text{null}} N_\tau \dot{q}$$

Opposes redundant joint velocity.
Zero at zero joint velocity.

d_{null} : damping coefficient [N m s/rad]

$\dot{q}$: measured joint velocity

- Conditioning torque

$$\tau_\sigma = k_\sigma N_\tau (s_\sigma v_7)$$

Seeks better local conditioning.
Can act at zero joint velocity.

k_σ : torque magnitude [N m]

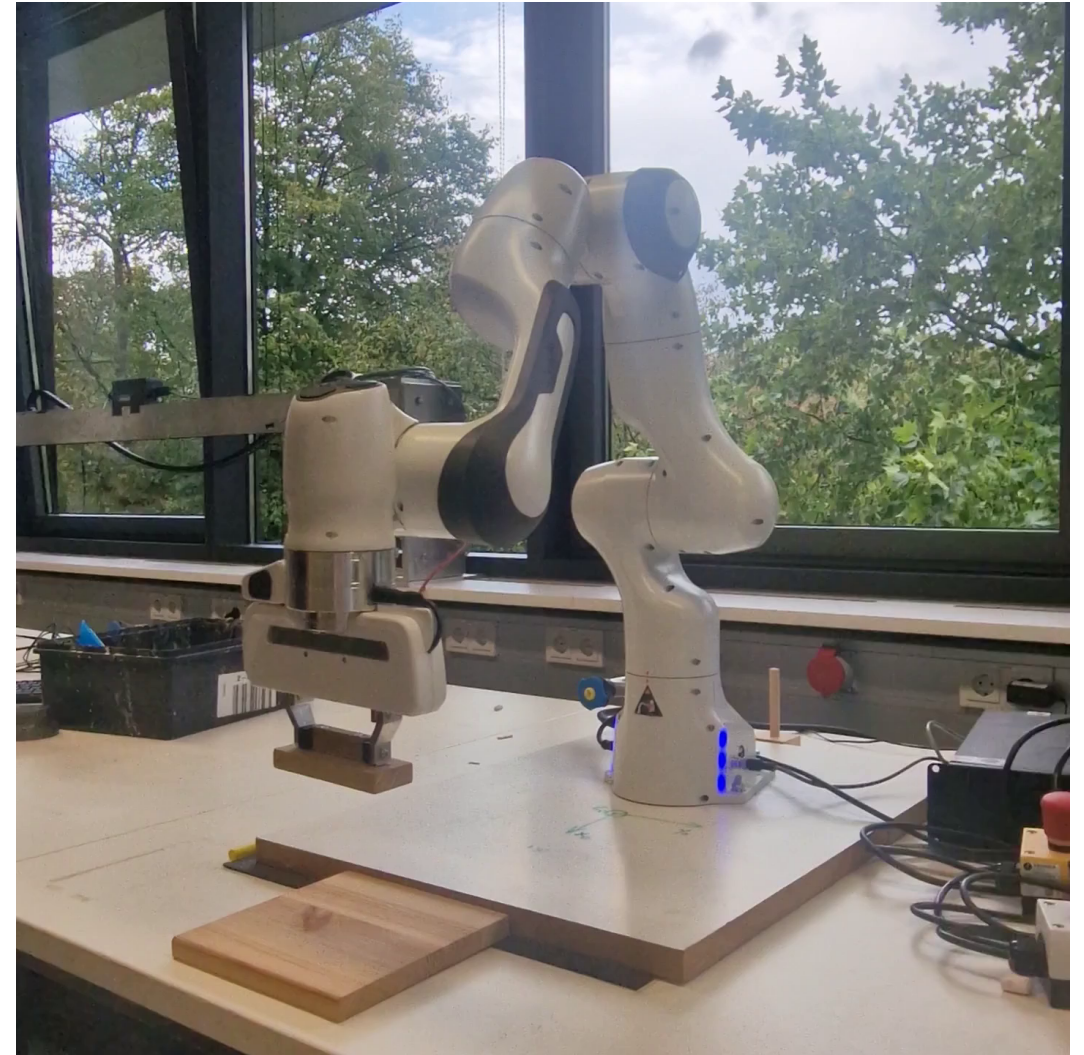
v_7 : unit null-space direction

s_σ : direction choice, zero in deadband

Disturbance demonstration

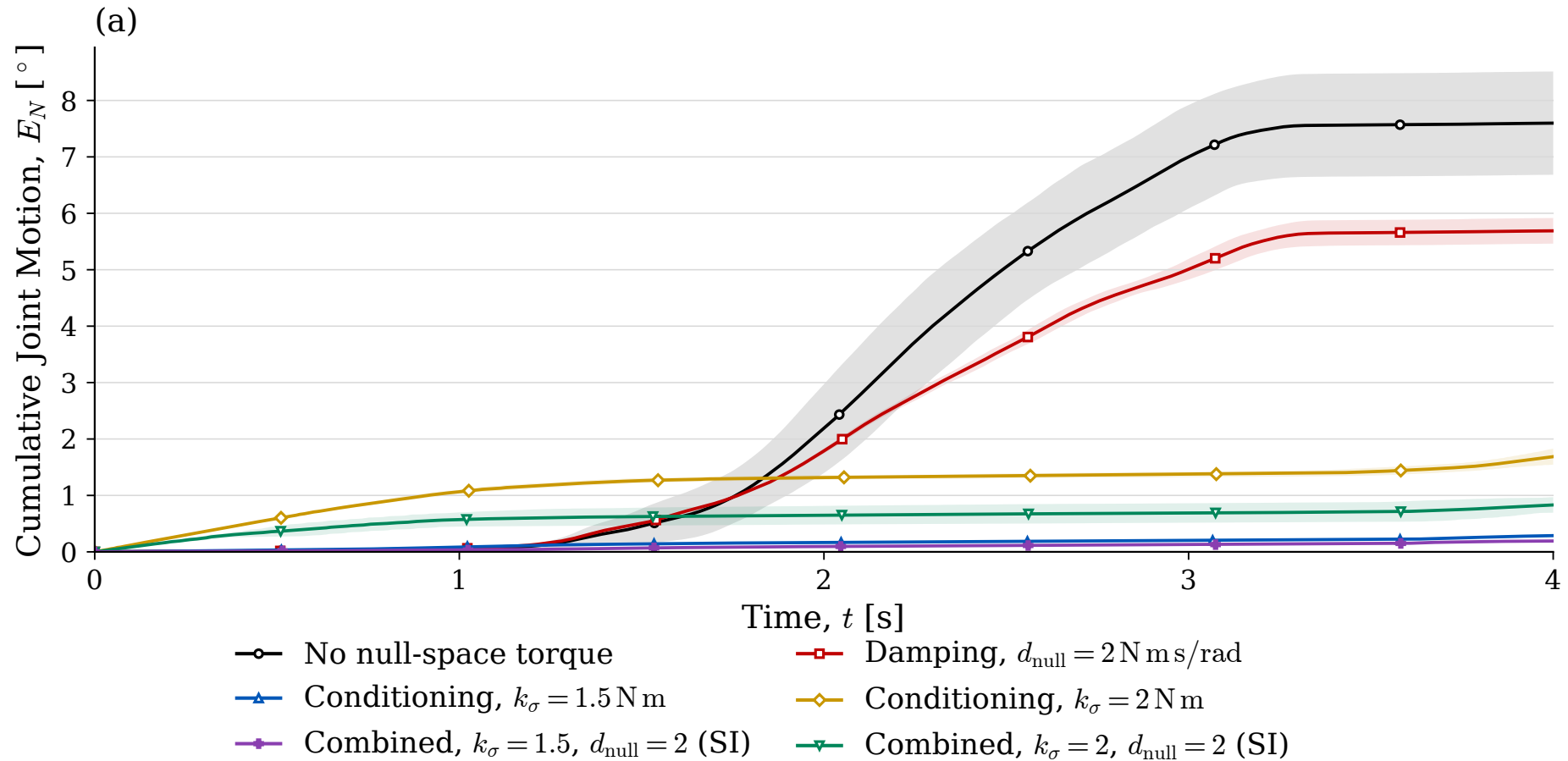


Null-space demonstration 1

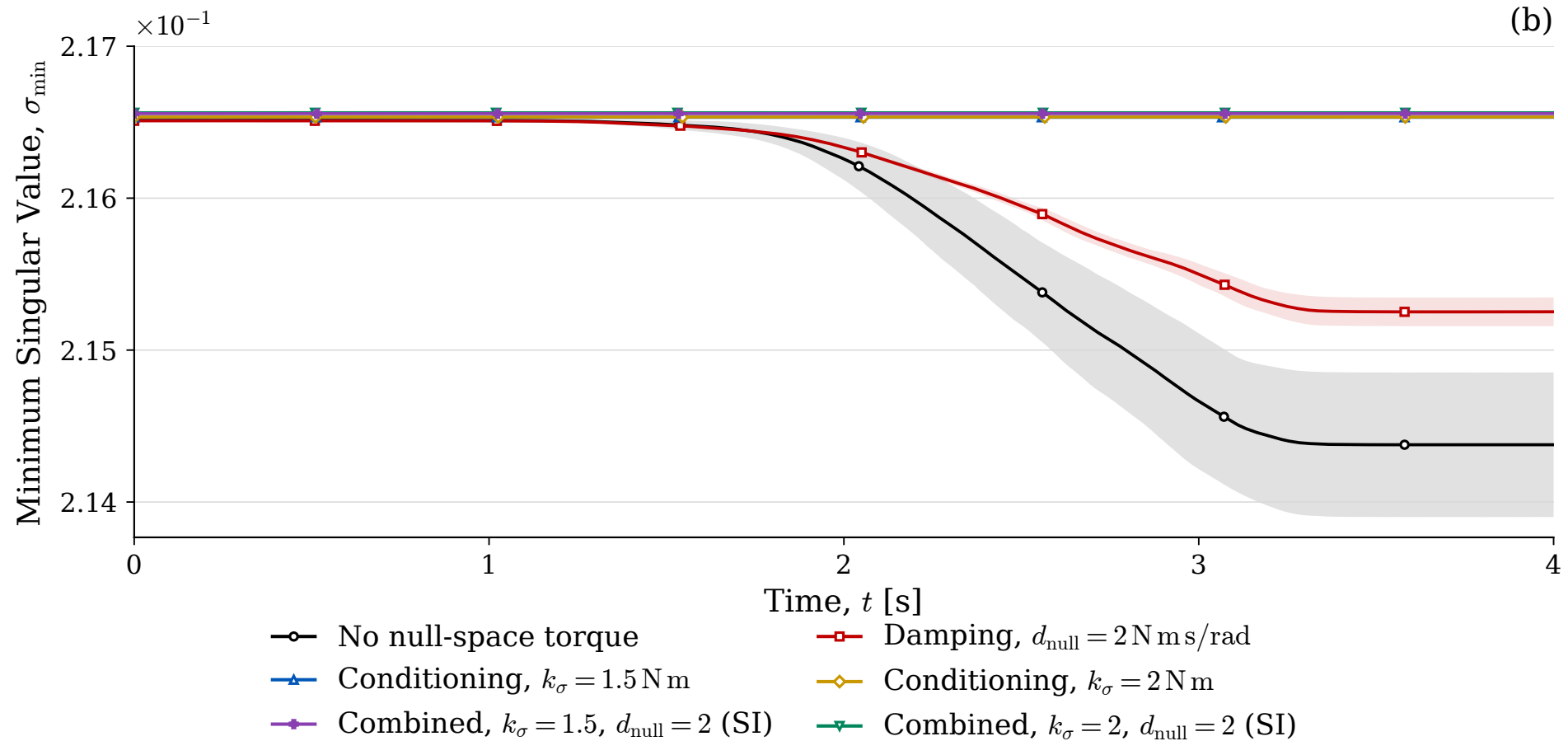


Null-space demonstration 2

Cumulative joint motion: all settings

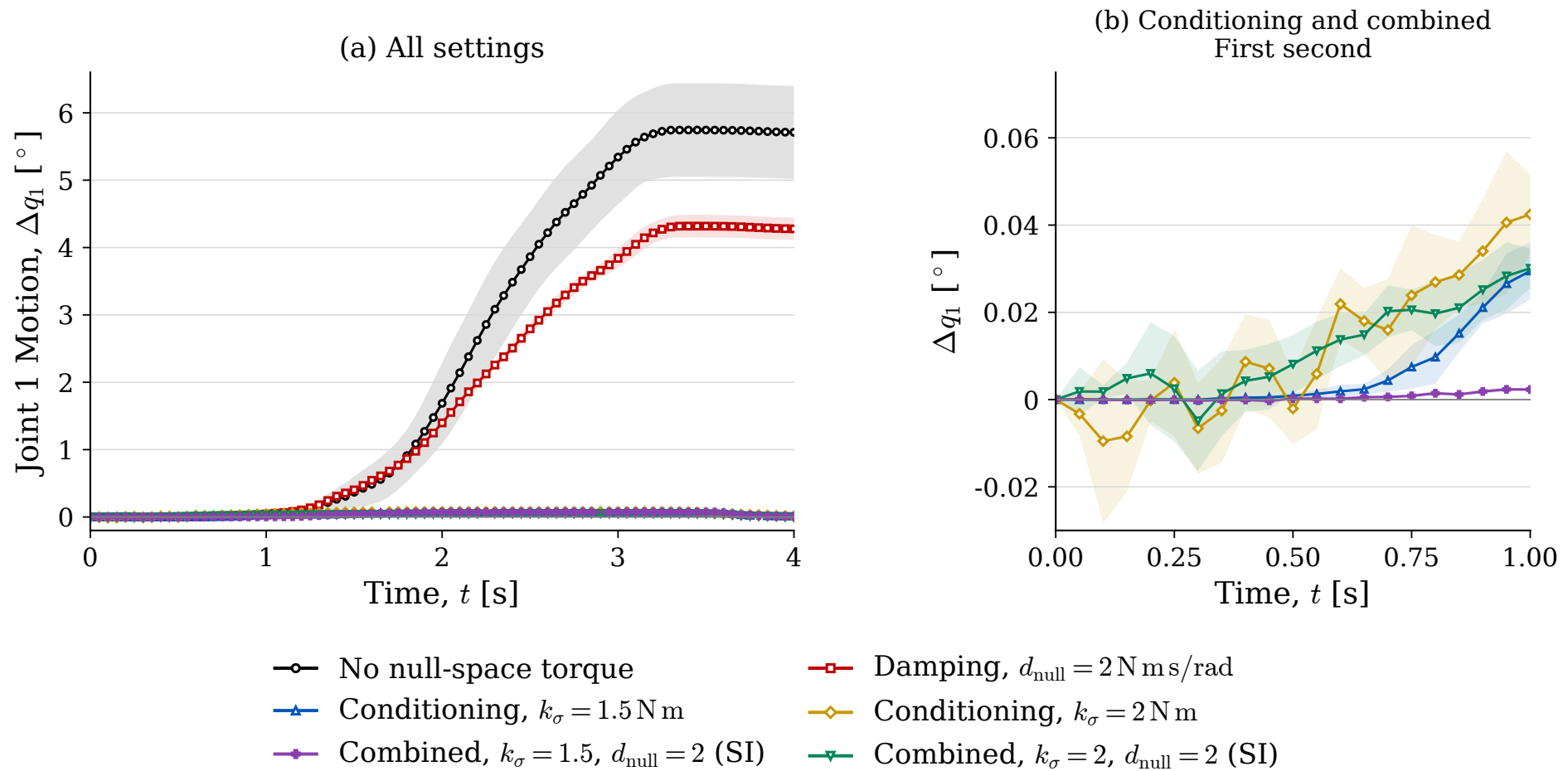


- Damping alone: 7.60° to 5.69°. Conditioning alone: 0.29° and 1.69°.
- Combined: 0.19° at 1.5 N m and 0.83° at 2 N m.



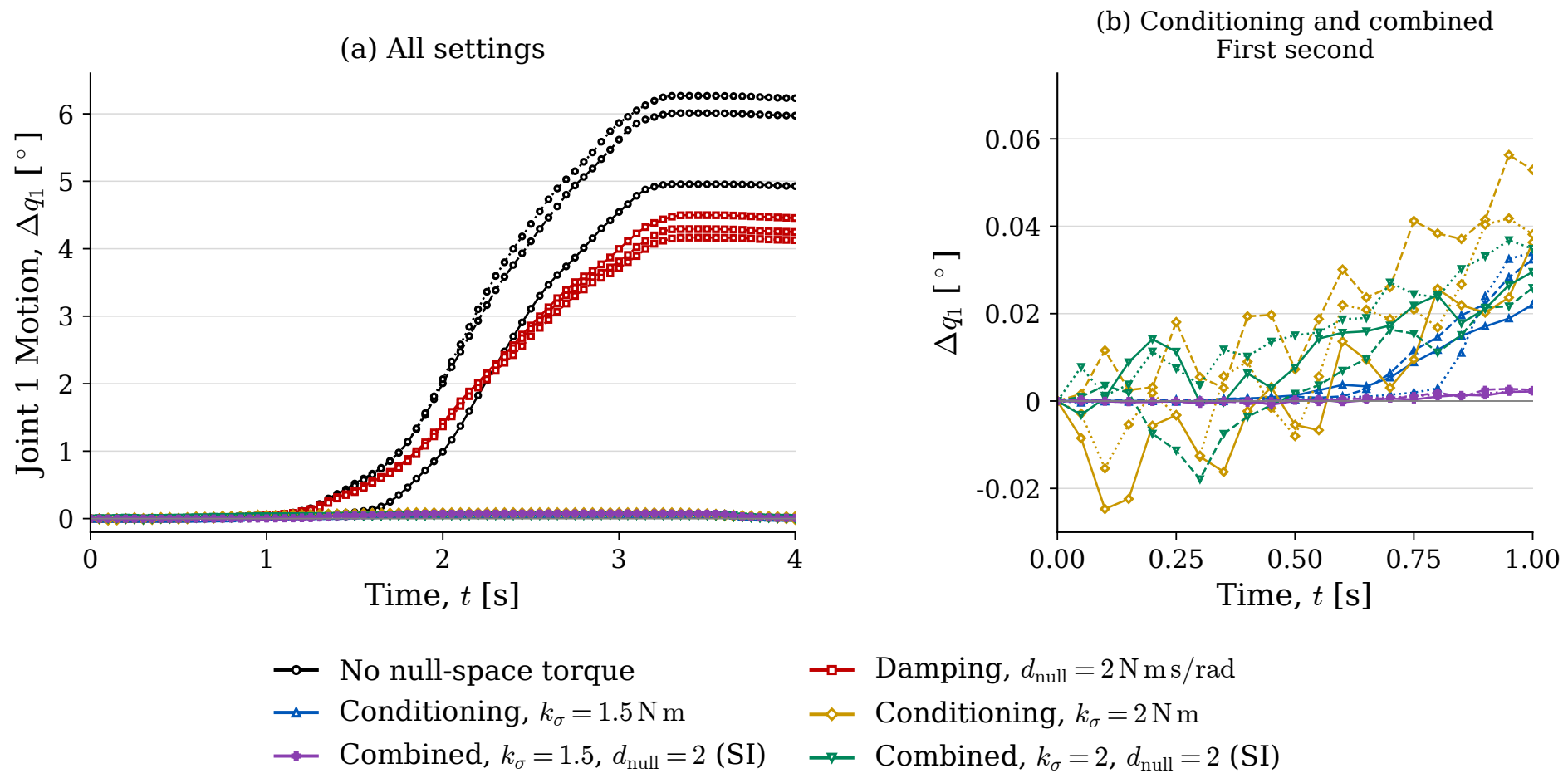
- Minimum singular value decreased with no torque and damping alone.
- Conditioning and combined control kept the indicator nearly constant.

Joint motion: mean of three trials



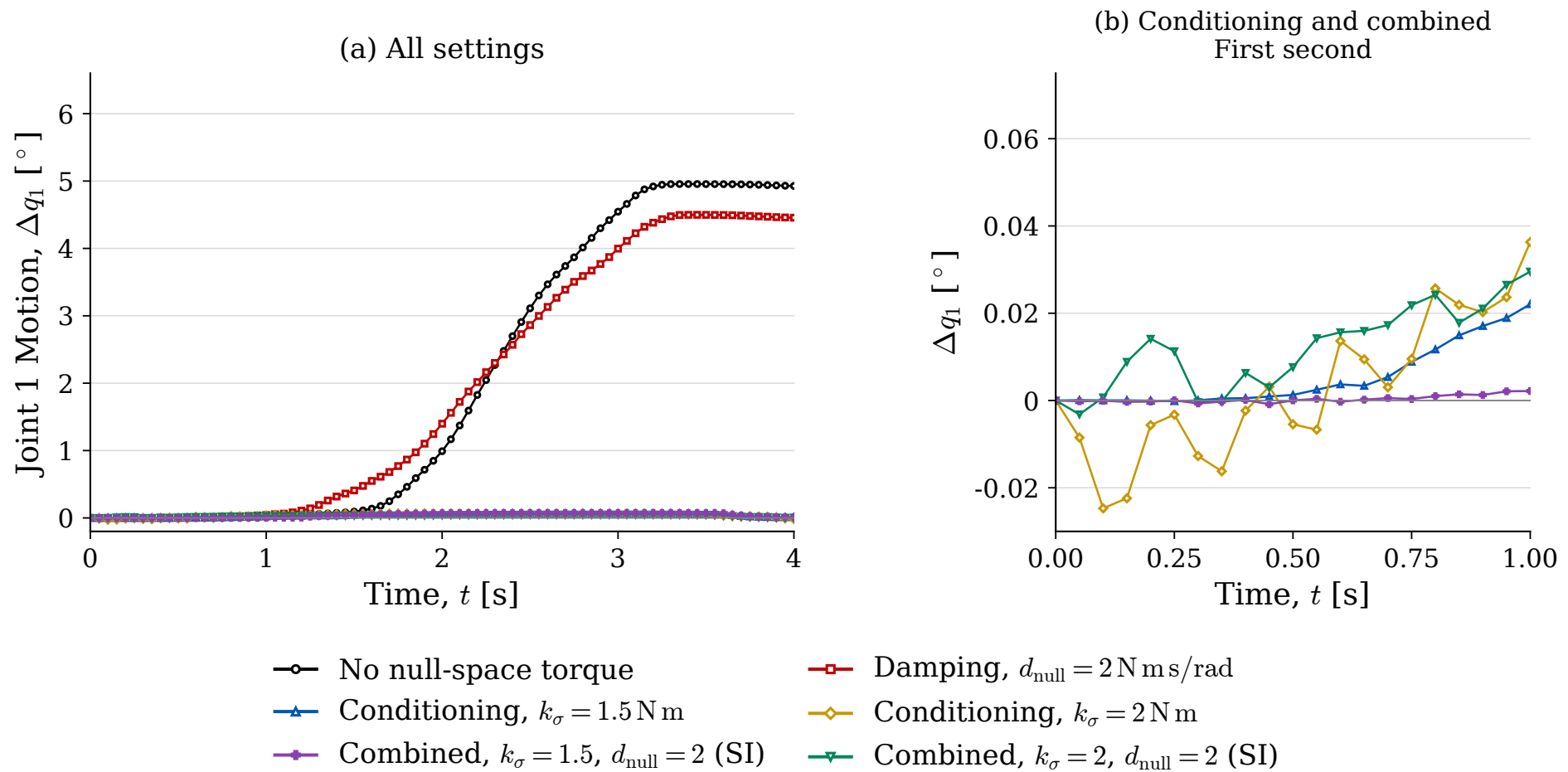
- Three-trial mean and one sample standard deviation.
- Joint 1 motion, including both combined settings.

Joint motion: all individual trials



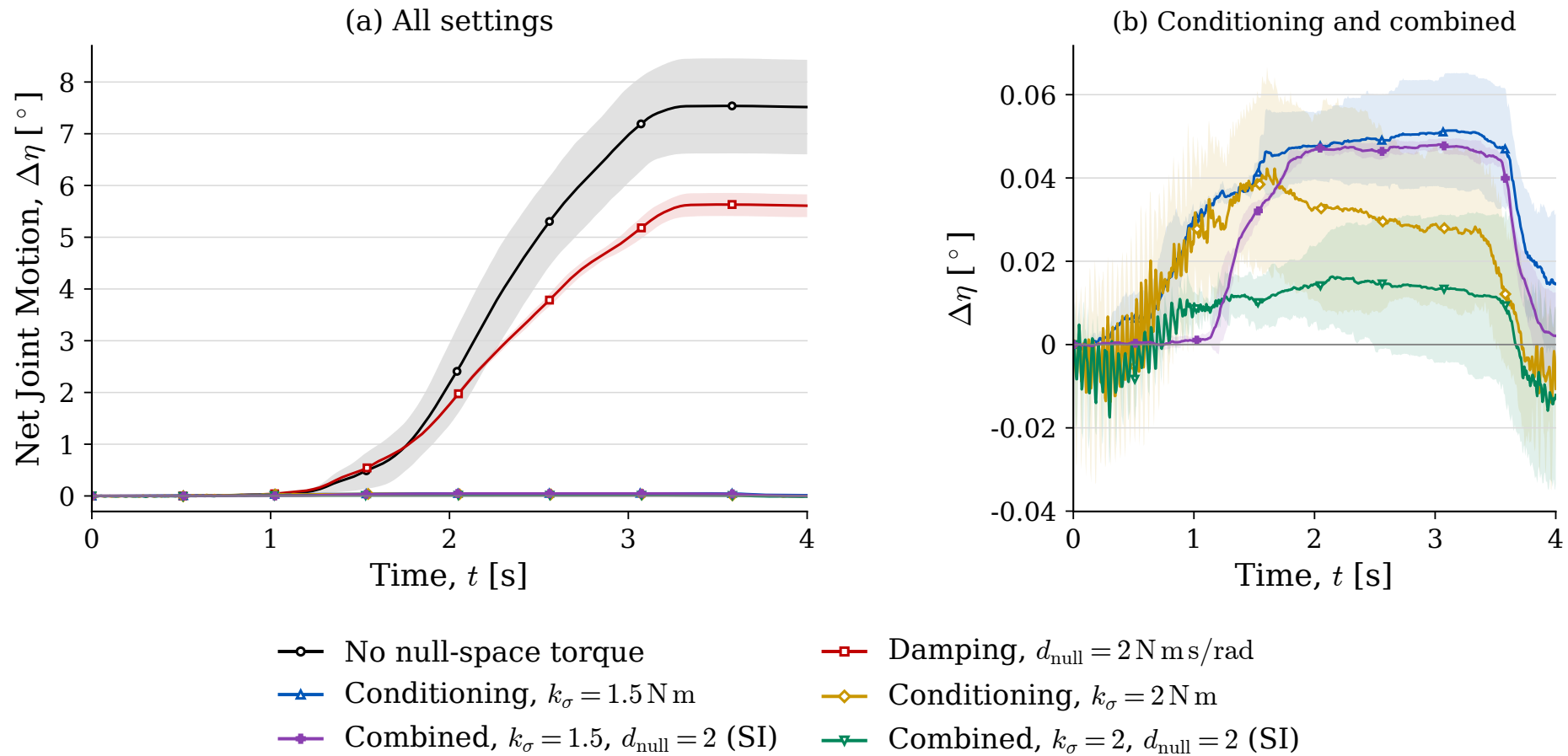
- Six settings, including conditioning with damping.
- Three individual trials per setting, each starting at zero.

Joint motion: one trial per setting



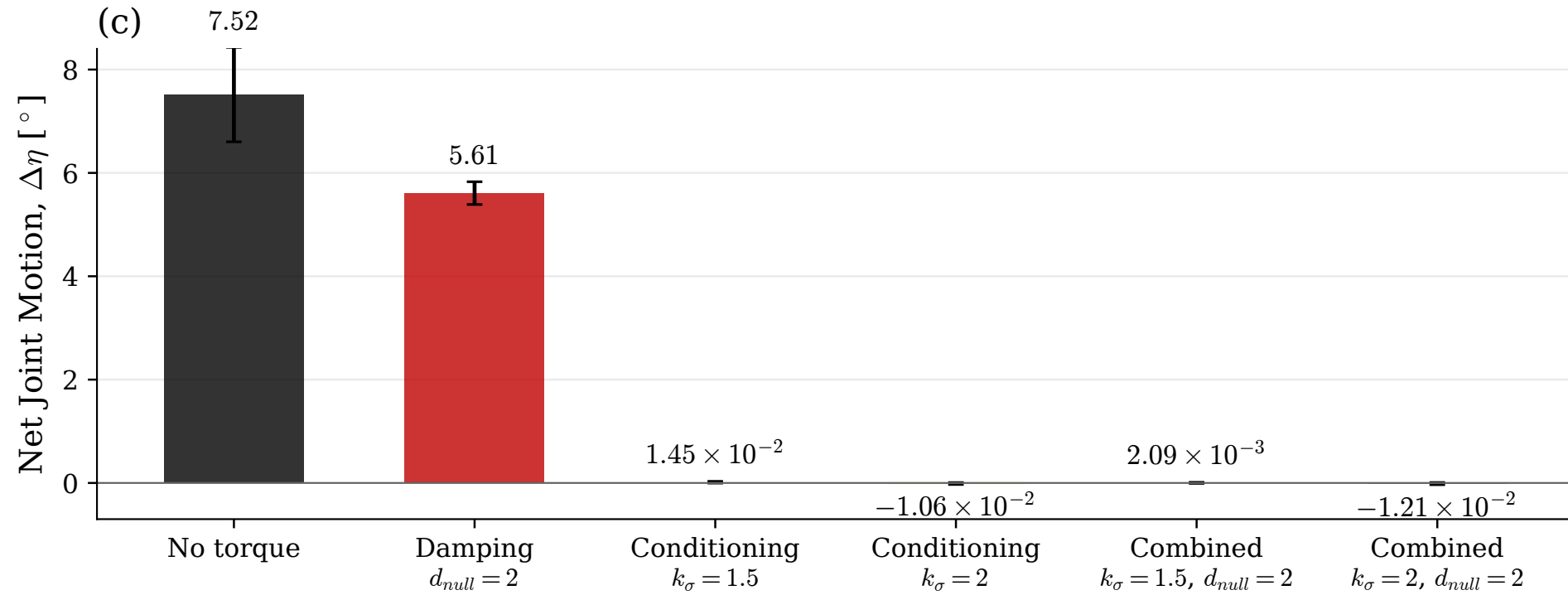
- Repetition 1 for each of the six settings.
- The close-up compares conditioning with and without damping.

Net joint motion over time



- Returning motion reduces the net joint motion.
- Conditioning and combined control end near zero.

Net joint motion



Three-trial mean and one sample standard deviation. Coefficients in SI units.

- Both combined settings left nearly zero mean net joint motion.
- Adding damping reduced cumulative motion at both conditioning settings.